# Lecture Notes on Numerical Linear Algebra and Data Science

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October 6, 2026

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# Chapter 1

# Linear Systems

## 1.1 Session 1: Matrices and Data

Matrices are a natural way to store data. A matrix

$$A = (a_{ij}) \in \mathbb{R}^{m\times n}$$

has $m$ rows and $n$ columns; $a_{ij}$ denotes the entry in row $i$ and column $j$. For example, a small data set could be recorded as

$$A = \begin{pmatrix} 90 & 150 & 0.2 & 1920 \\ 800 & 70 & 0.4 & 1988 \\ 150 & 90 & 1.3 & 2004 \end{pmatrix}.$$

Typically, rows represent data points and columns represent features. Thus, depending on the application, a row might represent an observation and a column its weight, age, or another measured quantity.

A vector $v \in \mathbb{R}^n$ is a matrix with one column,

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

Vectors are added componentwise and multiplied by scalars componentwise:

$$v + w = \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix}, \qquad \lambda v = \begin{pmatrix} \lambda v_1 \\ \vdots \\ \lambda v_n \end{pmatrix}.$$

**Definition 1.1.** *A* linear combination *of vectors $v_1, \dots, v_k$ is an expression*

$$\lambda_1 v_1 + \cdots + \lambda_k v_k, \qquad \lambda_1, \dots, \lambda_k \in \mathbb{R}.$$

For example,

$$2\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} - 3\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + 0\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \\ -14 \end{pmatrix}.$$

The coefficients may be arbitrary real numbers; a coefficient equal to zero simply means that the corresponding vector does not contribute.

For $A \in \mathbb{R}^{m\times n}$ with columns $A_1, \dots, A_n$ and $v = (v_1, \dots, v_n)^T \in \mathbb{R}^n$, matrix–vector multiplication is defined by

$$Av = v_1 A_1 + \cdots + v_n A_n.$$

For instance,

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}.$$

Equivalently, the $i$th entry of $Av$ is

$$(Av)_i = \sum_{j=1}^{n} a_{ij} v_j.$$

**Definition 1.2.** *The* column space *of $A \in \mathbb{R}^{m \times n}$ is*

$$\mathcal{C}(A) := \{Av : v \in \mathbb{R}^n\} = \operatorname{span}\{A_1, \ldots, A_n\} \subseteq \mathbb{R}^m.$$

Geometrically, a column space in $\mathbb{R}^3$ can be only the origin, a line through the origin, a plane through the origin, or all of $\mathbb{R}^3$. Which of these possibilities occurs depends on how many genuinely different directions are present among the columns. Three columns in one plane, for instance, can generate at most that plane.

**A recurring question.** Given feature vectors $A_1, \ldots, A_n \in \mathbb{R}^m$ and a target vector $b \in \mathbb{R}^m$, can $b$ be explained by the features? In algebraic form, the question is whether there are coefficients $x_1, \ldots, x_n$ such that

$$b = x_1 A_1 + \cdots + x_n A_n,$$

or, equivalently, whether $b \in \mathcal{C}(A)$. If the answer is yes, the problem is to find such coefficients. If the answer is no, we shall later look for a vector in $\mathcal{C}(A)$ that is close to $b$.

Adding a column which is already a linear combination of existing columns does not enlarge the column space. Equivalently, deleting a redundant column does not shrink it.

**Lemma 1.3.** *Let $A \in \mathbb{R}^{m \times n}$, and let $A'$ be obtained by appending a column $A_{n+1} \in \mathcal{C}(A)$. Then*

$$\mathcal{C}(A') = \mathcal{C}(A).$$

*Proof.* The inclusion $\mathcal{C}(A) \subseteq \mathcal{C}(A')$ is immediate. Conversely, write $A_{n+1} = \sum_{j=1}^{n} \mu_j A_j$. Every linear combination involving the new column can then be rewritten using only $A_1, \ldots, A_n$. □

More generally, if some $A_i$ is a linear combination of the remaining columns, then

$$\operatorname{span}\{A_1, \ldots, A_n\} = \operatorname{span}\{A_1, \ldots, A_{i-1}, A_{i+1}, \ldots, A_n\}.$$

Indeed, a coefficient multiplying $A_i$ can be distributed among the other columns. For example, if $A_3 = A_1 + A_2$, then

$$3A_1 + 5A_2 - 7A_3 = -4A_1 - 2A_2.$$

This is why redundant features may be deleted without changing the column space.

**Definition 1.4.** *Vectors $v_1, \ldots, v_k \in \mathbb{R}^m$ are* linearly dependent *if one of them can be written as a linear combination of the others. Otherwise they are* linearly independent.

Equivalently, $v_1, \ldots, v_k$ are linearly independent if and only if the only solution of

$$\lambda_1 v_1 + \cdots + \lambda_k v_k = \mathbf{0}$$

is the trivial solution $\lambda_1 = \cdots = \lambda_k = 0$. Hence a matrix $A$ has linearly independent columns precisely when $Ax = \mathbf{0}$ implies $x = \mathbf{0}$.

The criterion is often easiest to use through explicit relations. If, for example, a four-column matrix satisfies

$$A_3 = 3A_1 + 5A_2, \qquad A_4 = 4A_1 + 6A_2,$$

then its last two columns add no new directions: all four columns are generated by $A_1, A_2$. In contrast, when no nontrivial relation exists, the columns provide independent directions even if they live in a much larger ambient space.

**Definition 1.5.** *The* rank *of a matrix $A$ is the largest number of linearly independent columns of $A$.*

Thus $0 \le \operatorname{rank}(A) \le n$ for $A \in \mathbb{R}^{m\times n}$. We shall soon see the stronger bound $\operatorname{rank}(A) \le \min\{m, n\}$ and an algorithm for determining the rank.

**Remark 1.6.** *The number of columns is the number of available features; the rank is the number of independent directions represented by them. Adding an exact copy or an exact linear combination of existing features does not increase the rank.*

## 1.2 Session 2: Linear Systems and Elimination

Consider a linear system

$$Ax = b, \qquad A \in \mathbb{R}^{m\times n}, \quad b \in \mathbb{R}^m,$$

where $x \in \mathbb{R}^n$ is unknown. The system asks which linear combination of the columns of $A$ generates $b$. For example,

$$\begin{pmatrix} 2 & 1 & 4 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

is equivalent to

$$2x_1 + x_2 + 4x_3 = 1, \qquad x_1 + 3x_3 = -2.$$

We record the system in its augmented matrix $(A \mid b)$.

There are three elementary row operations:

1. swap two rows;
2. multiply a row by a nonzero scalar;
3. add a scalar multiple of one row to another row.

On an augmented matrix they act on the complete rows, including the right-hand side. Typical examples are

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 4 & 5 & 6 \\ 1 & 2 & 3 \end{array}\right),$$

$$\left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right) \xrightarrow{R_1 \leftarrow 2R_1} \left(\begin{array}{cc|c} 2 & 4 & 6 \\ 4 & 5 & 6 \end{array}\right), \qquad \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -3 & -6 \end{array}\right).$$

We write $A \sim B$ if $B$ can be obtained from $A$ by a sequence of elementary row operations. Row equivalence is symmetric because each elementary row operation has an inverse.

**Lemma 1.7.** *Suppose $(A \mid b) \sim (A' \mid b')$. Then $x_0$ solves $Ax = b$ if and only if it solves $A'x = b'$. In particular, elementary row operations preserve the solution set of a linear system.*

*Proof.* It is enough to check the three elementary operations. Swapping equations, multiplying an equation by a nonzero number, and replacing an equation by itself plus a multiple of another equation are all reversible and preserve exactly the same common solutions. □

Row operations generally change the column space. They do, however, preserve all linear relations among corresponding columns.

**Theorem 1.8.** *Let $A = (A_1, \dots, A_n)$ and $B = (B_1, \dots, B_n)$ with $A \sim B$. For every $\lambda_1, \dots, \lambda_n \in \mathbb{R}$,*

$$\lambda_1 A_1 + \cdots + \lambda_n A_n = \mathbf{0} \quad \Longleftrightarrow \quad \lambda_1 B_1 + \cdots + \lambda_n B_n = \mathbf{0}.$$

*Consequently,* $\operatorname{rank}(A) = \operatorname{rank}(B)$.

*Proof.* The coefficient vector $\lambda = (\lambda_1, \dots, \lambda_n)^T$ solves $A\lambda = \mathbf{0}$ exactly when it solves $B\lambda = \mathbf{0}$, because the two homogeneous systems are row equivalent. Thus a chosen subset of columns of $A$ is independent if and only if the corresponding subset of columns of $B$ is independent. □

**Definition 1.9.** *A matrix is in* row-echelon form *(REF) if*

1. *all nonzero rows occur above all zero rows; and*
2. *the first nonzero entry in every nonzero row occurs strictly to the right of the first nonzero entry in the row above it.*

*The first nonzero entry of a nonzero row is a* pivot*; its column is a* pivot column*.*

A matrix is in *reduced row-echelon form* (RREF) if, in addition, every pivot is 1 and every other entry in a pivot column is 0. Thus the pivot columns of an RREF are columns of the identity matrix. For example, a row-echelon matrix whose leading entries lie in columns $1, 3, 5$ has three pivots, irrespective of the values in its nonpivot columns.

**Theorem 1.10.** *Every matrix can be brought into row-echelon form, and indeed into reduced row-echelon form, by elementary row operations.*

*Proof.* This is the Gaussian elimination algorithm. In the leftmost column that is not yet processed, find a nonzero entry, swap its row into the current pivot position, and eliminate the entries below it. Continue on the remaining submatrix. This gives REF. Scaling pivots to 1 and eliminating above them gives RREF. □

As a larger elimination example, consider

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 12 & 1 \\ 0 & 4 & 8 & 2 & 12 & 12 \\ 0 & 8 & 16 & 4 & 6 & 21 \\ 0 & 2 & 4 & 1 & 3 & 5 \\ 0 & 6 & 12 & 3 & 21 & 13 \end{pmatrix}.$$

The first column is already zero and can never contain a pivot. In the remaining columns we first move a convenient nonzero row upward, use its leading entry to clear the entries below, and then repeat on the smaller lower-right block. Zero rows move to the bottom. The point of writing the algorithm this way is that no guessing is involved: at each stage we search for the next possible pivot and eliminate below it.

In RREF, every nonpivot column is visibly a linear combination of the pivot columns: its entries are exactly the coefficients of this combination. The pivot columns are distinct standard unit vectors and hence independent. Therefore the number of pivots is the rank. Because row operations preserve all relations between corresponding columns, the same pivot indices identify a maximal independent subset among the original columns.

**Corollary 1.11.** *If $B$ is a row-echelon form of $A$ with $r$ pivots, then*

$$\operatorname{rank}(A) = \operatorname{rank}(B) = r.$$

*In particular, for $A \in \mathbb{R}^{m\times n}$,*

$$\operatorname{rank}(A) \le \min\{m, n\}.$$

In particular, any collection of more than $m$ vectors in $\mathbb{R}^m$ is linearly dependent. Put the vectors into the columns of a matrix. Its rank is at most $m$, so not all of its more than $m$ columns can be independent. This is the formal version of the geometric fact that $\mathbb{R}^m$ contains only $m$ independent directions.

To obtain a maximal independent subset of the *original* columns of $A$, row-reduce $A$, record the pivot-column indices, and select the columns of $A$ with precisely those indices. One must not take the columns from the reduced matrix if the goal is a basis of $\mathcal{C}(A)$, since row operations can change the column space.